Classical Forecasting vs. Time-Series Foundation Models

Comparing established methods and pretrained models across 12 datasets

5 classical methods | **3 foundation-model families** | **12 datasets**

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Why this comparison matters

Do time-series foundation models consistently outperform classical forecasting approaches across datasets with different statistical characteristics?

Zero-shot promise

Can a pretrained model transfer across domains, frequencies, and horizons without dataset-specific training?

Model families

Compare that promise against established statistical, algorithmic, and neural forecasting approaches.

Different metrics

Separate point accuracy from quantile and predictive-distribution quality; they answer different questions.

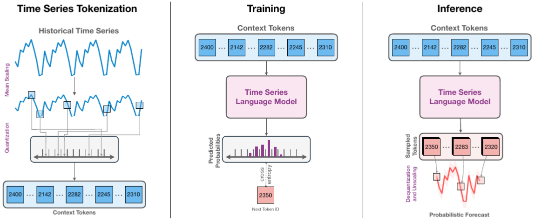
Statistical tests

Test whether observed performance differences are systematic or compatible with sampling variation.

Classical and conventional forecasting models

Model	Forecasting mechanism	Main strength	Main limitation
Seasonal Naive	Repeats the most recent seasonal cycle.	Transparent and robust reference for seasonal data.	Cannot adapt to trend or changing seasonal structure.
ARIMA $(2, 1, 2)$	Models differenced values through autoregressive and moving-average terms.	Captures short-range linear dynamics with a compact local model.	Fixed order; limited under nonlinear behavior or regime changes.
AutoARIMA	Searches candidate ARIMA orders and selects the best fit by AIC.	Automates model specification for each series.	Search can be costly or unstable on short and noisy histories.
Prophet	Decomposes a series into trend, changepoints, and calendar seasonalities.	Interpretable components and flexible trend changes.	A rigid additive decomposition may miss local dynamics.
DeepAR	Learns an autoregressive recurrent network under a probabilistic likelihood.	Represents nonlinear temporal relationships and uncertainty.	Requires training and hyperparameter choices for each task.

Chronos: forecasting as a language-model task



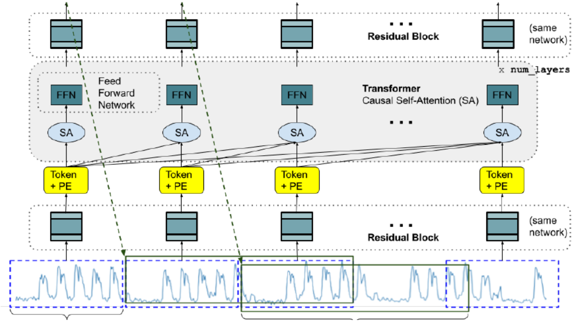
- **Representation:** mean-scale the series, quantize values into bins, and map them to a fixed token vocabulary.
- **Architecture:** a T5 encoder–decoder learns next-token probabilities using cross-entropy.
- **Forecast:** autoregressive token samples are dequantized into trajectories. Inference is **stochastic**; the sampled trajectories define a **probabilistic forecast**.

Chronos–T5 variants

Variant	Params	Enc./Dec.
Tiny	8M	4 / 4
Mini	20M	4 / 4
Small	46M	6 / 6
Base	200M	12 / 12
Large	710M	24 / 24

The variants use the same tokenization and forecasting mechanism. Capacity increases through greater depth and width.

TimesFM: patch the signal, decode the future



Patch tokens

Non-overlapping blocks of 32 observations are normalized and projected into token embeddings.

Decoder-only

Causal self-attention lets each patch representation use only its available history.

Multi-horizon

Each output head predicts a block of future time points; blocks are composed to reach the requested horizon.

Forecast behavior

TimesFM 2.5 directly outputs the mean and quantiles q10–q90. The values are **probabilistic forecasts**, but inference is **deterministic**: the same input returns the same distribution.

Moirai: one model for any frequency and variate count

Multiple patch sizes

A bank of input/output projections supports several patch lengths. The model can match its temporal resolution to series with different sampling frequencies and local dynamics.

Any-variate attention

Patches from every variable are flattened into one token sequence. Time and variable identifiers preserve where each patch came from, so attention can learn both within- and cross-variable dependence.

Distribution forecast

Masked future patches are decoded into an uncertainty representation. The original Moirai uses a flexible mixture density; Moirai 2.0 directly predicts quantiles. Inference is deterministic.

Twelve datasets I: competition and archive panels

Dataset	Domain	Full dataset	Series used	Freq. / H	Observations	Statistical characteristics
M1 Yearly	Business / economic	181 univariate series	20	Y / 6	15–58	Trends, level shifts, weak or absent seasonality
M3 Quarterly	Business / economic	756 univariate series	20	Q / 8	24–72 points	Quarterly cycles, trend, nonstationarity
M4 Hourly	Mixed competition	414 univariate series	20	h / 48	748–1,008 points	High frequency, daily seasonality, noise
Weather (rain)	Environment	Rain subset of 3,010 series	20	D / 30	1,332–65,981	Zero inflation, intermittency, seasonal weather
Car Parts	Retail / inventory	2,674 univariate series	20	M / 12	51 each	Sparse intermittent demand, many zeros
NN5 Weekly	Banking / ATM	111 univariate series	20	W / 8	about 113	Annual seasonality, holidays, demand volatility

Twelve datasets II: health, infrastructure, and macroeconomics

Dataset	Domain	Full dataset	Series used	Freq. / H	Observations	Statistical characteristics
National Illness	Public health	1 series, 7 variables	1 target	W / 24	966 weeks	Epidemic peaks, annual cycle, regime change
Traffic	Transport	1 series, 862 sensors	5 targets	h / 24	17,544 hours	Strong daily/weekly cycles, correlated sensors
PSM	Server telemetry	1 series, 25 variables	5 targets	h / 24	132,481 min. rows, resampled h	Anomalies, abrupt shifts, high-frequency noise
Elexon Demand	Energy	1 national demand series	1	D / 30	2024–2026 (730 days)	Weekly cycle, weather/calendar effects, nonstationarity
USGS Streamflow	Hydrology	5 gauge series	5	D / 30	2015–2026 ($\sim$4,230 days)	Annual hydrology, skew, floods/droughts, gaps
BLS Macro	Macro	3 indicator series	3	M / 6	2010–2026 ($\sim$199 months)	Trends, structural breaks, heterogeneous scales

Point and probabilistic forecast evaluation

MASE point

$$\mathrm{MASE} = \frac{H^{-1}\sum_{h=1}^{H}|y_h - \hat{y}_h|}{(T-m)^{-1}\sum_{t=m+1}^{T}|y_t - y_{t-m}|}$$

Error scaled by an in-sample seasonal-naive error. It is comparable across units and datasets.

< 1 beats the scale; $= 1$ matches it; > 1 is worse.

WQL quantiles

$$\mathrm{WQL} = \frac{1}{|Q|}\sum_{q\in Q}\frac{2\sum_h \rho_q(y_h - \hat{y}_{q,h})}{\sum_h |y_h|}$$

Weighted quantile (pinball) loss on $Q = \{0.1, \ldots, 0.9\}$, normalized by target magnitude.

Multiple quantiles reveal asymmetric risk and interval quality.

CRPS distribution

$$\mathrm{CRPS}(F, y) = \int_{-\infty}^{\infty} \left(F(z) - \mathbf{1}\{y \le z\}\right)^2 dz$$

Scores the entire predictive CDF against the observation; rewards **calibration** and **sharpness** jointly.

Implemented as a finite-quantile approximation from q10–q90.

Central forecast
How accurate is the median path?

Quantile accuracy
Where are under- and over-predictions?

Distribution quality
How well does the predictive CDF match reality?

Together they separate scale-free point error from quantile and distributional quality; lower is better.

Hypothesis tests and confidence sets for predictive ability

Diebold–Mariano

$d_t = L_t(A) - L_t(B)$
$H_0: \mathbb{E}[d_t] = 0$

Tests equal predictive accuracy for two models while estimating the sampling variance of the serially dependent loss differential.

Insight: direction and significance of one pairwise difference.

SPA

$d_{k,t} = L_t(B) - L_t(k)$
$H_0: \max_k \mathbb{E}[d_{k,t}] \leq 0$

Tests whether any candidate improves on benchmark B. Bootstrap critical values account for searching over many alternatives.

Insight: evidence that the benchmark is genuinely outperformed.

Model Confidence Set

$H_0: \mathbb{E}[d_{ij,t}] = 0$ for all $i, j \in \mathcal{M}$

Begins with all models and sequentially removes the worst-supported member until equal predictive ability is no longer rejected.

Insight: $\widehat{\mathcal{M}}_{1-\alpha}$, the set of models statistically compatible with best performance.